NOTATION · CHEAT SHEET

Linear System Notation at a Glance

The marks from the notation cards across the linear-system pages: how each is read and what it means. Every group links to the full card on its page.

GAUSSIAN ELIMINATION

$[A \mid \mathbf{b}]$	**THE AUGMENTED MATRIX** Coefficients left of the bar, constants right of it.
$R_i + cR_j \to R_i$	**ROW I PLUS C TIMES ROW J, INTO ROW I** The arrow points at the row being replaced.
$A \sim B$	**A IS ROW-EQUIVALENT TO B** Same solutions, different entries: never write $=$.

ECHELON FORM

REF, RREF	**ROW ECHELON FORM; REDUCED** REF is not unique; RREF is.
$\boxed{2}$	**A BOXED PIVOT** Marks the leading nonzero entry of its row.
$\mathbf{p} + t\mathbf{v}_1 + s\mathbf{v}_2$	**THE GENERAL SOLUTION** A particular solution plus one free direction per free column.

HOMOGENEOUS SYSTEMS

$A\mathbf{x} = \mathbf{0}$	**A X EQUALS THE ZERO VECTOR** Bold $\mathbf{0}$: a column of zeros, not the number 0.
$\mathbf{x} = \mathbf{0}$	**THE TRIVIAL SOLUTION** Always a solution of a homogeneous system.
$\mathrm{Nul}(A)$	**THE NULL SPACE OF A** The whole solution set of $A\mathbf{x} = \mathbf{0}$.

SOLVABILITY

consistent	**CONSISTENT OR INCONSISTENT** At least one solution, or none at all.
$\mathrm{rank}[A \mid \mathbf{b}]$	**RANK OF THE AUGMENTED MATRIX** Equal to $\mathrm{rank}(A)$ exactly when the system is consistent.
$0,\ 1,\ \infty$	**NONE, ONE, OR INFINITELY MANY** The only possible counts of solutions.

SAME MARKS, DIFFERENT JOB

$\sim$	Row equivalence here; on the basis-change page, similar matrices.
$\mathbf{0}$ vs 0	The zero vector against the number zero.
the bar in $[A \mid \mathbf{b}]$	A separator, not divides as in number theory.